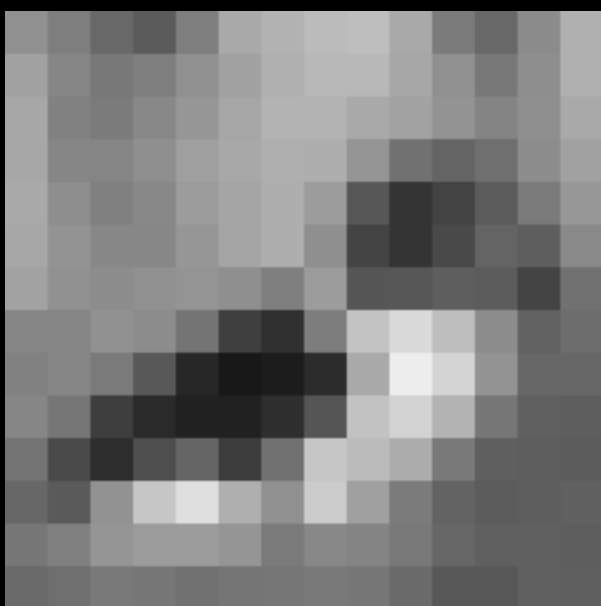
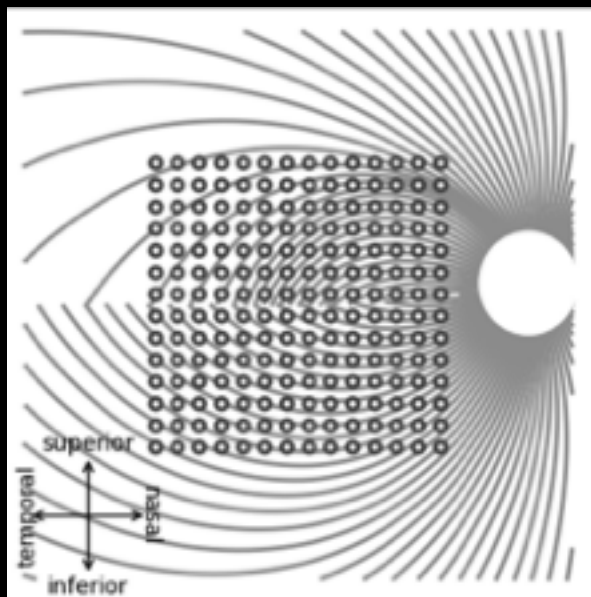


Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

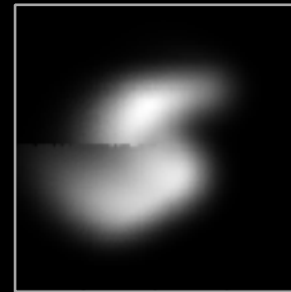
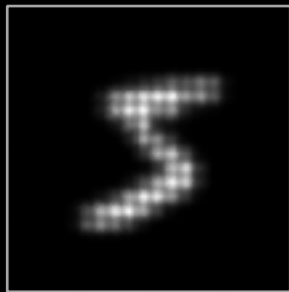
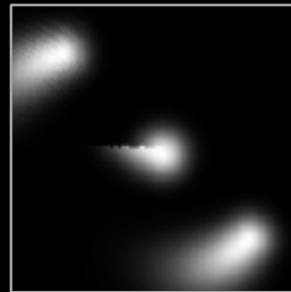
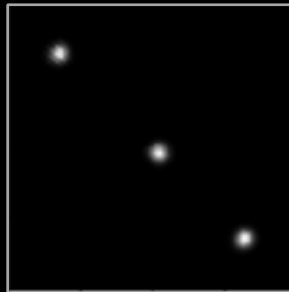
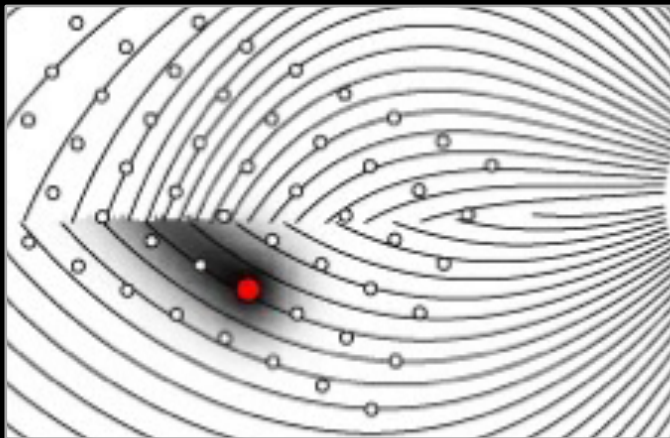
RWTHAACHEN
UNIVERSITY



Limitations and Challenges



Limited
Electrode
Resolution

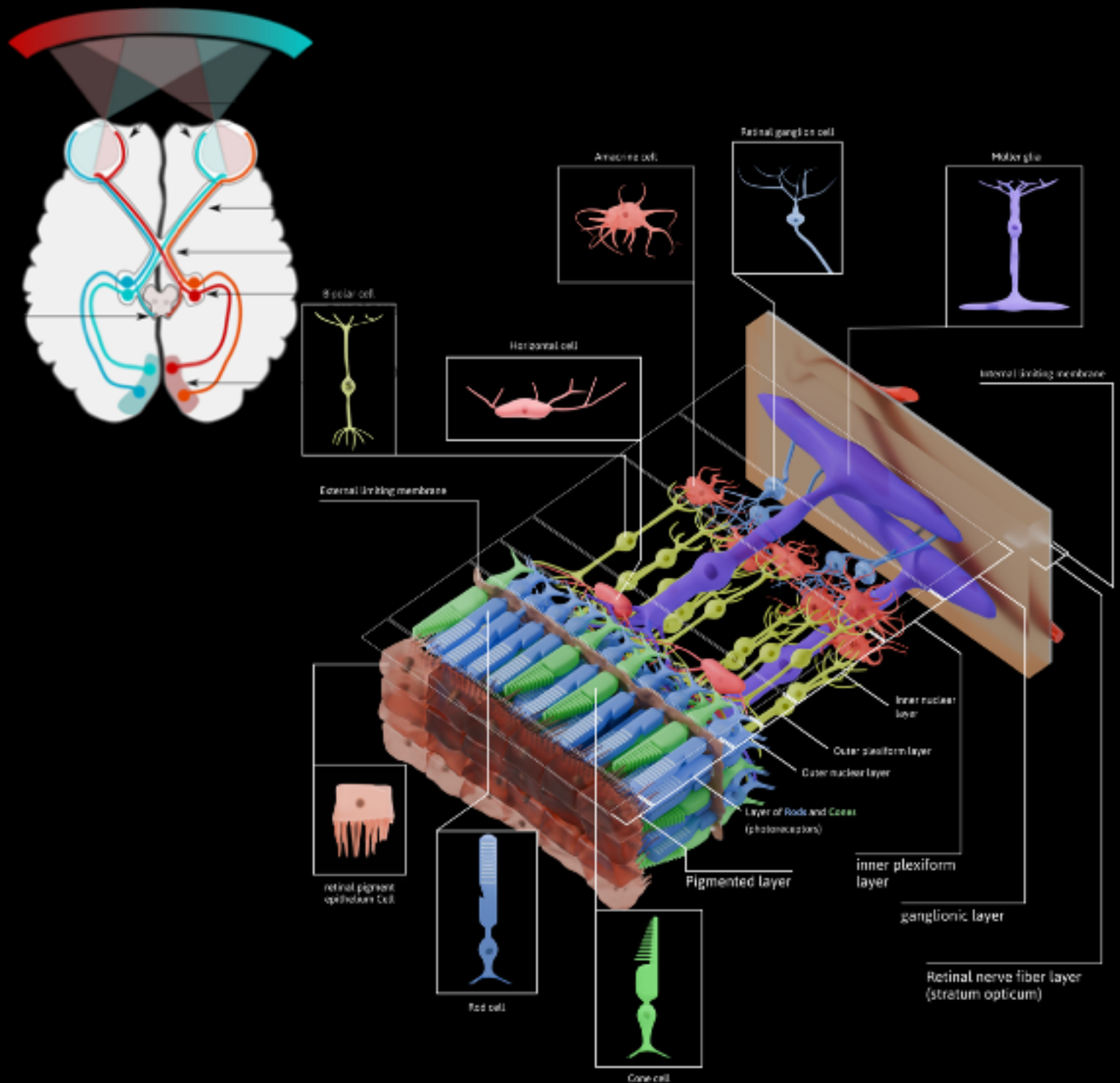


Distortion

- *350 patients with Argus II*
- *38 patients with PRIMA*
- *4 patients for pulse2percept*

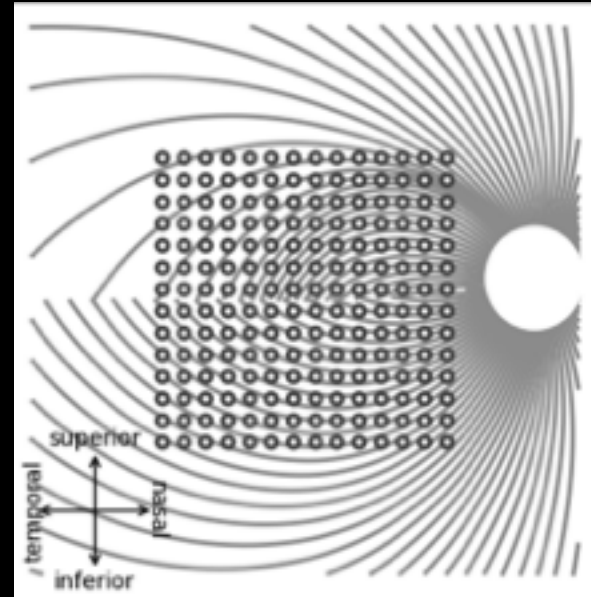
Limited Patients' Data

Miquel Perello Nieto; Otosakanum2.
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Complex Neural Circuitry

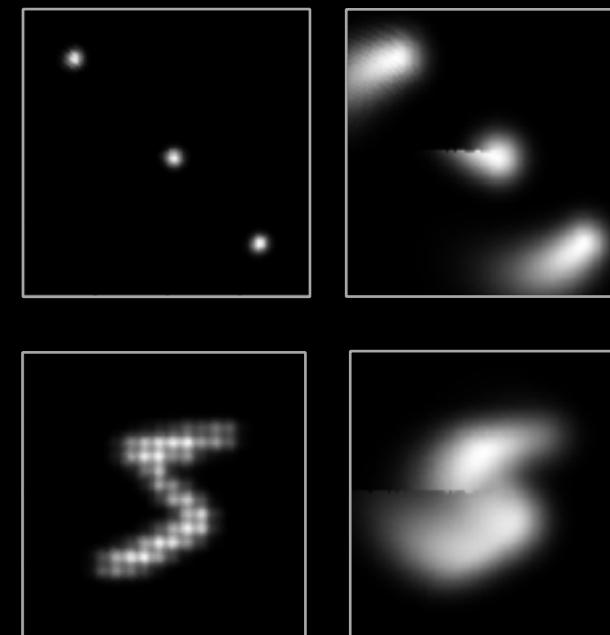
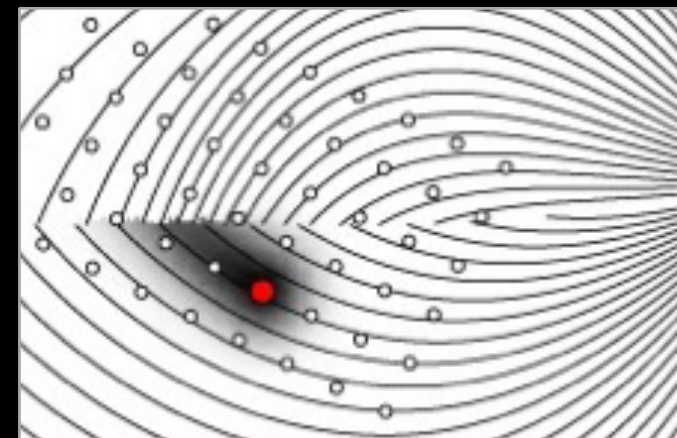
Limitations and Challenges



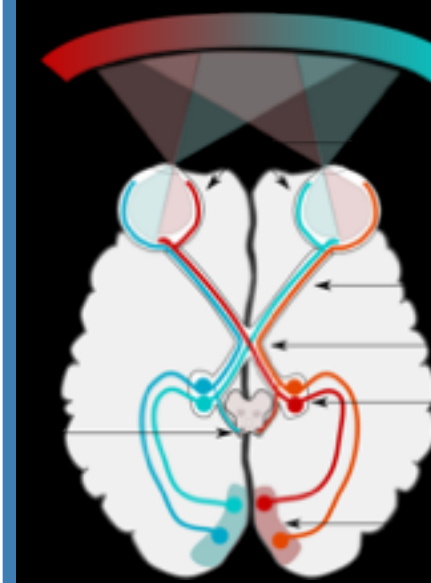
Limited
Electrode
Resolution

- 350 patients with Argus II
- 38 patients with PRIMA
- 4 patients for pulse2percept

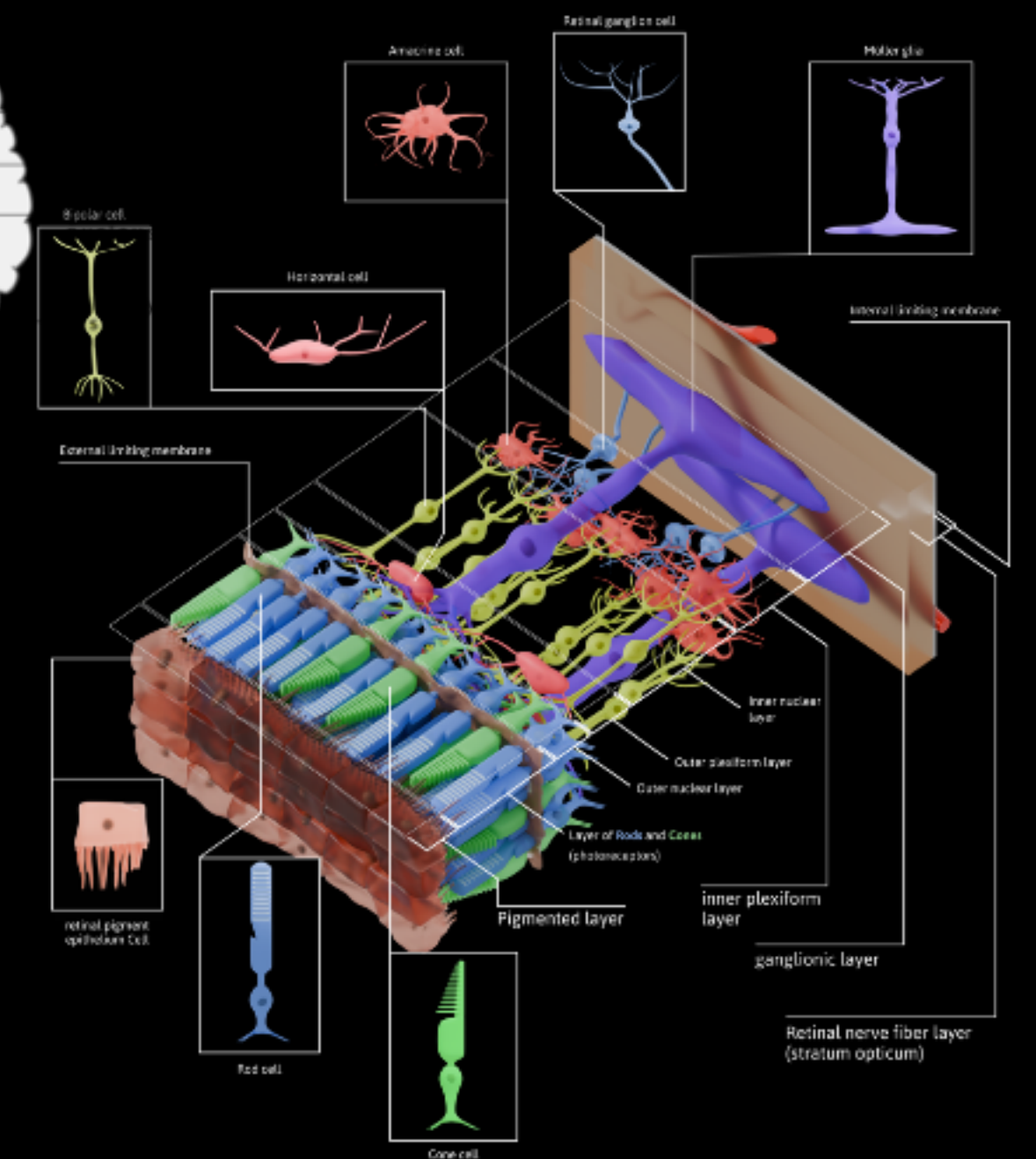
Limited Patients' Data



Distortion

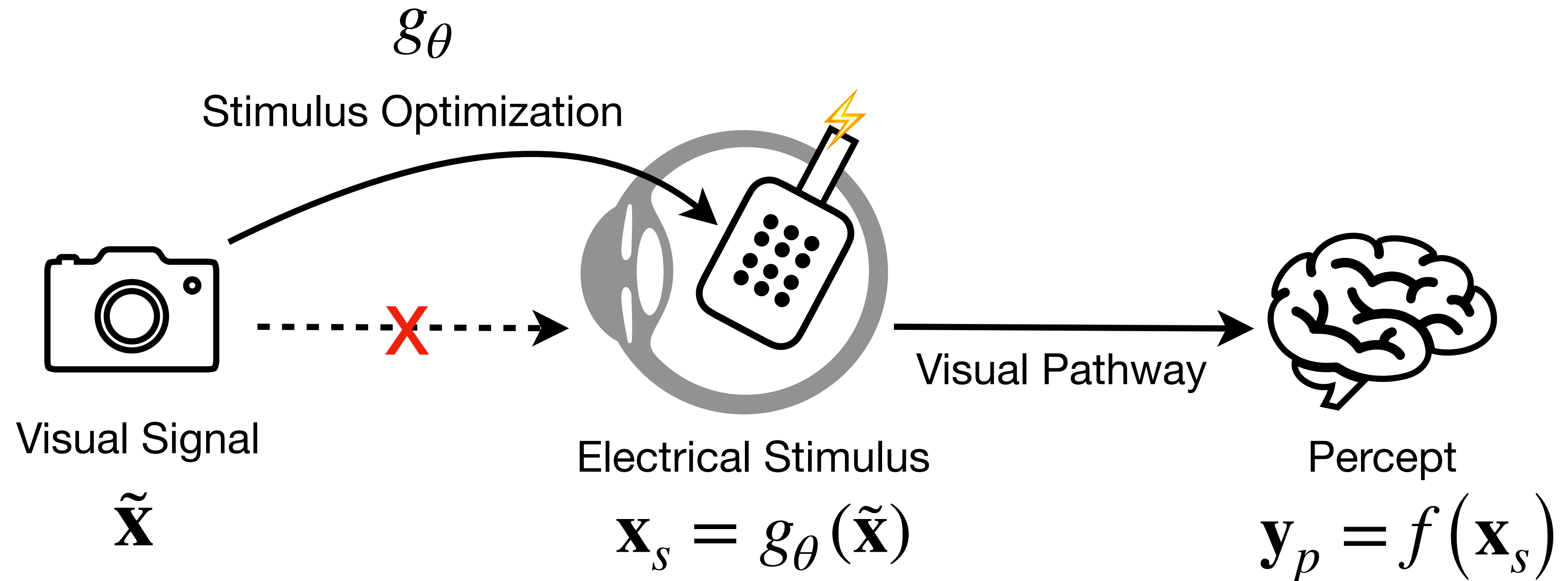


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Complex Neural Circuitry

Problem Formulation



$$\arg \min_{\theta} \mathcal{L}_{eval} \left(\tilde{\mathbf{x}}, f_{\phi} \left(g_{\theta}(\tilde{\mathbf{x}}) \right) \right)$$